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Natural Sciences Manual

Grades 7-9

Chapter 1

Natural Sciences Manual: Grades 7 to 9

Thematic Organization

This manual consolidates the Natural Sciences content for grades 7, 8, and 9, organized by scientific domains rather than by school year. This thematic approach facilitates understanding the relationships between different concepts and natural processes.

Content adapted from: O Bichinho do Saber (www.obichinhodosaber.com)

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- Geological landscapes
- Internal structure of the Earth
- Earth's subsystems

1.2.2 Rocks and Minerals

- Rocks and minerals
- Sedimentary rocks
- Igneous and metamorphic rocks
- Rock cycle

1.2.3 Earth's Dynamics

- Continental Drift Theory
- Plate Tectonics Theory
- Folds and Faults
- Volcanism
- Earthquakes

1.2.4 Earth's History

- Fossils and rock dating
- Stages of Earth's History

1.2.5 Geological Sustainability

- Sustainable exploitation of rocks
- The contribution of geological knowledge

1.3 CELL BIOLOGY AND ECOLOGY

1.3.1 Conditions for Life

- Earth conditions that allow the existence of life

1.3.2 Cell Organization

- The microscope
- The cell and levels of biological organization

1.3.3 Ecosystems

- Levels of ecosystem organization
- Abiotic factors
- Biotic factors
- Food chains and matter cycles
- Ecological successions

1.4 SUSTAINABILITY AND ENVIRONMENTAL MANAGEMENT

1.4.1 Sustainable Development

- Sustainable development
- Natural resources

1.4.2 Environmental Protection

- Catastrophes
- Measures for ecosystem protection
- Nature protection and conservation

1.4.3 Territory and Resource Management

- Land use planning and management
- Waste and water management

1.5 HUMAN BIOLOGY

1.5.1 Health and Body Organization

- Individual and community health
- Structural levels of the human body

1.5.2 Nutrition and Digestion

- Foods, nutrients, and healthy eating
- Digestive system

1.5.3 Circulatory and Immune System

- Blood
- Cardiovascular system
- Lymphatic system

1.5.4 Respiratory System

- Respiratory system
- Basic life support

1.5.5 Excretory System

- Excretory system: urinary system
- Excretory system: skin

1.5.6 Coordination and Regulation

- Neuro-hormonal system: nervous system
- Neuro-hormonal system: hormonal system

1.5.7 Reproduction and Heredity

- Reproductive system
- Heredity

Chapter 2

GEOLOGY

2.1 Landscapes and Earth's Structure

2.1.1 Geological landscapes

2.1.1.1 FUNDAMENTAL DEFINITIONS

Landscape "Extension of territory that can be observed at a given location up to the horizon line, consisting of natural and/or human elements"

Outcrop "Portion of rock emerging from the soil surface"

Geological Landscape "Landscape whose predominant element is rock"

2.1.1.2 CLASSIFICATION OF GEOLOGICAL LANDSCAPES

Plutonic Igneous Landscapes - Characteristic rock: granite - Landforms: granitic massifs, block chaos

Volcanic Igneous Landscapes - Characteristic rock: basalt - Landforms: volcanic cones, calderas, basaltic columns

Metamorphic Landscapes - Deformed rocks (quartzites, schists) - Many with foliation (organization in layers) - Forms: folds, faults, quartzitic ridges

Detrital Sedimentary Landscapes - Predominate sedimentary aggregates - Examples: beaches, dunes, ravines, cliffs, fairy chimneys

Chemical Sedimentary Landscapes - Limestone or evaporites (precipitated by evaporation) - Examples: lapiaz fields, sinkholes, caves, swallow holes, rock salt mines

2.1.1.3 DISTRIBUTION IN PORTUGAL

Landscape Type	Location
Plutonic igneous	North and center
Volcanic igneous	Azores and Madeira
Metamorphic	North, center, and Alentejo
Sedimentary	Western and southern margins, Tagus and Sado basins

Essential Learning: Characterize the school surroundings landscape through dominant rocks and relief, using field data.

2.1.2 Internal structure of the Earth

2.1.2.1 Study of Earth's internal structure

Methods used in studying Earth's internal structure

Direct methods:

- Allow collection of material from Earth's interior for analysis
- Only explore the most superficial part
- Include observation of rock outcrops
- Observation of materials expelled by volcanoes
- Excavations and mining explorations
- Geological drilling

Indirect methods:

- Obtain information about the entire interior without collecting materials
- Astrogeology: study of meteorites reveals chemical composition of Earth's interior
- Seismology: seismic waves behave differently depending on density, distinguishing layers
- Geomagnetism: originates at the center, informing about structure and composition

Models of Earth's internal structure

Geochemical model:

- Based on chemical composition of materials
- Oceanic crust (basaltic rocks) and continental crust (granitic rocks)
- Mantle (iron and magnesium)
- Core (iron and nickel)

Geophysical model:

- Based on physical properties of materials
- Lithosphere (solid)
- Asthenosphere (plastic)

- Mesosphere (solid)
- Endosphere
  - Outer core (liquid)
  - Inner core (solid)

Essential Learning: Relate volcanic and seismic phenomena with direct/indirect methods and their importance for knowledge of internal structure, explaining contributions of science and technology.

2.1.3 Earth's Subsystems – Earth as a system

2.1.3.1 Earth's Subsystems

Earth functions as an integrated system composed of four interconnected subsystems:

- **Geosphere** (solid rocky layer)
- **Hydrosphere** (water in solid and liquid states)
- **Atmosphere** (gaseous envelope layer)
- **Biosphere** (living organisms)

2.1.3.2 System Interactions

Examples of interactions between subsystems: - Photosynthetic organisms remove carbon dioxide and release oxygen - Water evaporation cycles - Volcanic gas emissions - Erosive action of water on rocks

2.1.3.3 Origin of Life Theories

Four main hypotheses about the origin of life:

1. **Creationism** - divine creation
2. **Cosmozoic Hypothesis** - life from space
3. **Spontaneous Generation** - spontaneous generation
4. **Oparin's Evolutionary Hypothesis** - organic molecules formed from atmospheric compounds through electrical discharges

2.1.3.4 Urey-Miller Experiment

In 1953, Harold Urey and Stanley Miller built a device that attempted to recreate conditions when life emerged, successfully creating organic compounds without living organisms.

2.1.3.5 Current Theory

Life emerged from "protolife" (organic molecules) formed from methane and ammonia in hydrothermal vents on the ocean floor, followed by an evolutionary explosion 600 million years ago.

2.1.3.6 Soil Formation & Importance

Soils develop through slow weathering of rocks, organized in horizons (O, A, B, C layers), serving as: - Habitat - Support structure - Water retention - Nutrient fixation

2.1.3.7 Subsystem Roles

Each subsystem maintains life through specific functions: - **Geosphere** provides substrate and magnetic protection - **Hydrosphere** offers habitat and water reserves - **Atmosphere** provides essential gases and radiation protection - **Biosphere** provides organic matter

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2.2 Rocks and Minerals

2.2.1 Rocks and minerals

2.2.1.1 Geology and Petrology

Geology - science that studies the Earth

Petrology - branch of geology that studies rocks

2.2.1.2 Rock Classification

Rocks are classified into three main types:

1. **Igneous rocks**
   - Plutonic/intrusive
   - Volcanic/extrusive
2. **Metamorphic rocks**
3. **Sedimentary rocks**
   - Detrital
   - Chemical
   - Biogenic

Rocks are composed of different minerals. For example, granite is composed of various minerals.

2.2.1.3 Mineral Definition

A mineral is a **substance:** - Solid - Natural - Inorganic - With defined chemical composition - With crystalline structure

2.2.1.4 Mineral Properties for Identification

Minerals can be identified through eight main characteristics:

1. **Color** - visual appearance of the mineral
2. **Streak** (powder color) - color of the mineral when reduced to powder
3. **Luster** - appearance of the mineral's surface when it reflects light
   - Metallic
   - Pearly
   - Vitreous (glass)
4. **Cleavage/Fracture** - mineral breaking patterns
   - Cleavage: flat breaking surfaces
   - Fracture: irregular breaking surfaces
5. **Hardness** - resistance a mineral offers to scratching
   - Measured by Mohs Scale
6. **Specific properties**
   - Taste
   - Smell
   - Magnetism
   - Effervescence

2.2.1.5 Mohs Scale

The Mohs Scale is used as a practical tool for identifying minerals through hardness.

Essential Learning: The content emphasizes practical identification of minerals using observable properties and understanding that rocks are composed of different minerals.

2.2.2 Sedimentary rocks

2.2.2.1 Agents of external geodynamics

Agents that shape landscapes:

Water:

- Physical action: dissolves minerals, removes and transports rock fragments, freezes causing disintegration
- Chemical action: reacts with minerals causing alteration

Wind:

- Physical action: transports particles that collide with rock outcrops causing wear

Living organisms:

- Physical action: roots and burrowing fragment rocks
- Chemical action: release substances that react with rocks

2.2.2.2 Formation of sedimentary rocks

Sedimentogenesis (sediment formation):

- Weathering
- Erosion
- Transport
- Sedimentation

Diagenesis (rock consolidation):

- Compaction
- Cementation

Changes in sediments during fluvial transport:

"Heavier grains are deposited and left behind (remain near the source)" while "lighter grains are transported further and become more rounded"

Classification of sedimentary rocks:

Detrital (or clastic) sedimentary rocks - Formed from fragments of pre-existing rocks - Quartz and clay are common minerals - Types: claystone, sandstone, breccia, conglomerate

Chemical sedimentary rocks - Formed by precipitation of dissolved constituents - Limestone (calcite precipitation) - Evaporites: rock salt, gypsum

Biogenic sedimentary rocks - Formed from remains of living organisms - Fossiliferous limestone (reef, shell) - Coal

2.2.2.3 Sedimentary landscapes

Detrital sedimentary landscapes:

- Beaches and dunes
- Ravines
- Fairy chimneys
- Pedunculated blocks

Chemical sedimentary landscapes:

- Karstic landscapes (with sinkholes and caves)
- Rock salt mines

Essential Learning:

- Relate action of external geodynamic agents with landscape modeling
- Interpret water course dynamics models

- Explain sedimentary rock formation processes
- Distinguish detrital, chemical, and biogenic rocks

2.2.3 Igneous and metamorphic rocks

2.2.3.1 Formation of Igneous Rocks

Igneous (or magmatic) rocks form from the cooling of magma, either inside the Earth or at the surface.

Classification of Igneous Rocks:

Plutonic (intrusive) rocks - **Formation process:** Solidification at depth with slow cooling - **Texture:** Granular with visible crystals - **Examples:** Granite, Gabbro

Volcanic (extrusive) rocks - **Formation process:** Solidification at surface with rapid cooling - **Texture:** Aphanitic or glassy (microscopic or absent crystals) - **Examples:** Basalt, Rhyolite, Obsidian

2.2.3.2 Formation of Metamorphic Rocks

Metamorphic rocks originate from other rocks (sedimentary, igneous, or other metamorphic) that have been subjected to conditions of **high pressure and temperature**.

Types of Metamorphism:

Contact Metamorphism - **Conditions:** Occurs near magmatic intrusions - **Texture:** Non-foliated - **Rock examples:** Marble, Quartzite

Regional Metamorphism - **Conditions:** Occurs at convergent plate boundaries - **Texture:** Foliated (mineral orientation in layers) - **Rock examples:** Schist, Gneiss

2.2.3.3 Geological Landscapes

Different igneous and metamorphic rocks create characteristic geological landscapes:

Plutonic Landscapes:

- Granitic massifs
- Prismatic basalt columns

Volcanic Landscapes:

- Volcanic cones
- Calderas
- Volcanic caves

Metamorphic Landscapes:

- Quartzitic ridges

Learning Objectives:

- Distinguish different types of rocks (plutonic igneous, volcanic igneous, metamorphic)
- Understand the formation processes of each rock type
- Identify geological landscapes associated with each rock type

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2.2.4 Rock Cycle

2.2.4.1 Interaction between various types of rocks

Geological processes involved in the lithological cycle:

Rocks undergo continuous transformations through specific processes:

- **Rock melting** → generates magma
- **Magma cooling** → results in igneous rocks
- **Metamorphism** → produces metamorphic rocks
- **Sedimentogenesis and diagenesis** → originate sedimentary rocks
- **Rock outcropping** → occurs through tectonic uplift combined with erosion of upper layers
- **Rock subsidence** → happens through pressure exerted by layers above and tectonic movements

Central Concept of the Rock Cycle:

"Rocks are not finished products as they interact with each other, dynamically, in a cycle of continuous renewal in which they are formed, transformed, and destroyed."

This process results from **Earth's internal and external dynamics**.

Essential Learning:

- Interpret information related to the rock cycle, integrating knowledge about sedimentary, igneous, and metamorphic rocks
- Relate different types of rocks to Earth's internal and external dynamics

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2.3 Earth's Dynamics

2.3.1 Continental Drift Theory

2.3.1.1 Alfred Wegener's Continental Drift Theory

What the Continental Drift Theory proposes:

- The continents were once united as one (Pangaea), surrounded by a single ocean (Panthalassa)

- This supercontinent fragmented, originating Laurasia and Gondwana, which in turn also fragmented giving rise to current continents
- Meanwhile, continents have moved over time, floating like icebergs, to their current positions

Justification for continental movement:

- Earth's rotational force
- Attraction force of the Sun and Moon
- Tidal movement

2.3.1.2 Arguments in favor of this theory

Morphological arguments:

- Continents fit together like a puzzle

Paleontological arguments:

- Existence of fossils of the same age on different continents

Geological (or lithological) arguments:

- Geological continuities at continent edges

Paleoclimatic arguments:

- Existence of glacial traces in tropical zones and fossils of tropical plants in cold climates

2.3.1.3 Arguments against

Why this theory was not accepted at the time:

- It was demonstrated that Earth's rotational force and the attraction force of the Sun and Moon were not sufficient to cause continental movement
- Materials constituting continents do not allow them to float on ocean floors

Essential Learning: Systematize information about Continental Drift Theory, explaining arguments that supported it and that weakened it, considering its historical context.

2.3.2 Plate Tectonics Theory

2.3.2.1 Ocean floors

Ocean floor morphology:

- **Continental shelf**: relatively flat and shallow surface near continents
- **Continental slope**: slope between continental shelf and abyssal plain
- **Abyssal plain**: flat region of great depth
- **Ocean ridge**: mountain chain in the middle of the ocean floor

- **Rift**: valley located in the central region of the ocean ridge through which magma is expelled
- **Ocean trench**: deep and elongated depression

Age and paleomagnetism of ocean floor rocks:

- The closer to the rift, the more recent the rocks
- Rocks show symmetrical and parallel paleomagnetic record on both sides of the ridge

Ocean expansion:

- Harry Hess's theory about seafloor spreading: formed from rifts, growing symmetrically
- Contradicts Wegener by showing continents don't float on oceans

Exploration technology: sonar, autonomous underwater vehicles, remotely operated vehicles, sample collectors

2.3.2.2 Tectonic plates

Earth's surface layers:

- **Lithosphere**: most superficial layer, composed of solid and rigid rocks
- **Asthenosphere**: layer under the lithosphere, composed of molten rocks

Plate Tectonics Theory:

- Lithosphere fragmented into plates that move on the asthenosphere
- Movement originated by convection currents caused by internal heat
- Earth maintains volume through formation at rifts compensated by destruction in subduction zones

Tectonic plate boundaries:

- **Divergent**: plates move apart at rifts, forming lithosphere, with volcanism and seismicity
- **Convergent**: plates collide, destroying lithosphere in subduction zones or forming mountains
- **Transform**: plates slide over each other, without formation or destruction, with high seismicity

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2.3.3 Folds and Faults

2.3.3.1 Rock Deformation

Rock deformation occurs due to tectonic forces, which can be: - Compressive - Tensional - Shear (or transform)

Factors affecting rock deformation type include: - Type and intensity of force - Rock type - Temperature and pressure

2.3.3.2 Folds

Folds are caused by compressive forces and occur in rocks with ductile (plastic) behavior.

2.3.3.3 Faults

Faults result from compressive, tensional, or shear forces and develop in rocks with brittle (rigid) behavior.

Fault Classification:

1. **Normal Fault** - caused by tensional forces; the hanging wall block descends relative to the footwall block
2. **Reverse Fault** - caused by compressive forces; the hanging wall block rises relative to the footwall block
3. **Transform Fault** (strike-slip) - caused by shear forces; blocks move horizontally in opposite directions

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2.3.4 Volcanism

2.3.4.1 Volcanic Activity

Fundamental Definitions:

- **Volcanology**: "branch of Geology that studies volcanism"
- **Volcanologist**: "scientist who studies volcanism"
- **Volcanism**: "set of processes related to the release of materials from magma, from Earth's interior to the surface"

Types of Volcanism:

- **Primary volcanism**: manifests through volcanic eruptions (fissure type or central type)
- **Secondary volcanism**: manifests through emission of gases or water at high temperatures

2.3.4.2 Primary Volcanism

Composition of a Central Type Volcano:

- **Magma chamber**: underground reservoir where magma is found
- **Chimney**: through which magma ascends
- **Crater**: opening at the top through which volcanic materials are expelled
- **Volcanic cone**: conical elevation resulting from magma solidification

Active and Inactive Volcanoes:

- **Active**: "is erupting, or has potential to do so, or showed activity less than 10,000 years ago"
- **Inactive**: "has not shown activity for more than 10,000 years"

Magma versus Lava:

- **Magma**: "mixture consisting of molten rocks, gases, and some rock constituents in solid state, inside the Earth"
- **Lava**: "liquid material formed from magma when it releases its gaseous constituents, at the surface"

Types of Lava:

- **Scoriaceous lava**: flows slowly (viscous), irregular surface
- **Ropy lava**: flows easily (fluid), wrinkled appearance
- **Pillow lava**: surface cools rapidly in water, rounded surface

Materials Expelled by Volcanoes:

Liquid Materials (Lava): - Scoriaceous lava - Ropy lava - Pillow lava

Solid Materials (Pyroclasts): - Ash - Lapilli - Blocks - Bombs

Gaseous Materials: - Water vapor - Carbon dioxide (CO_2) - Sulfur dioxide (SO_2) - Carbon monoxide (CO)

Types of Volcanic Activity:

1. Explosive Activity:

- Magma rich in silica
- Viscous lava
- Many gases and pyroclasts
- Glowing clouds
- High and steep cone

2. Effusive Activity:

- Magma poor in silica
- Fluid lava
- Few gases
- Large lava flows
- Low and gently sloping cone

3. Mixed Activity:

- Alternation between effusive and explosive periods
- High but gently sloping cone

2.3.4.3 Secondary Volcanism

Secondary Manifestations:

- **Thermal springs**: "waters heated at depth that originate springs of mineral-rich waters"
- **Fumaroles**: "release of water vapor and other gases, at very high temperatures"

- **Geysers**: "intermittent and periodic jets of liquid water and water vapor"

2.3.4.4 Consequences of Volcanism

Benefits of Volcanism:

- Field fertilization
- Construction materials
- Energy production (geothermal)
- Thermalism
- Tourist landscapes
- Scientific contributions

Risks of Volcanism:

- Infrastructure destruction
- Injuries and deaths
- Climate changes
- Earthquakes
- Disruption of human activities

Volcanic Activity Monitoring:

- Seismic recording
- Surface deformation
- Temperature measurement
- Gas analysis
- Lahar detection
- Volcanic risk maps (allow emergency plans to be drawn up)

Volcano Locations:

- Pacific Ring of Fire
- Mid-Atlantic Ridge
- Mediterranean Belt
- Mainly at lithospheric plate boundaries

2.3.5 Earthquakes

2.3.5.1 Seismic activity

Seismology and earthquakes:

- **Seismology:** science that studies earthquakes
- **Earthquakes:** "sudden vibrations of the Earth's crust, at a given place, at a given time"

Foreshocks and aftershocks:

- **Foreshocks:** tremors before the main shock
- **Aftershocks:** tremors after the main shock

Causes of an earthquake:

- Magma displacement
- Landslides
- Cave collapse
- Human activities (explosions)
- Tectonic plate movement (main cause)

Tectonic earthquakes:

- **Hypocenter:** location inside the Earth where earthquake energy is released
- **Epicenter:** "point on the surface, vertically above the hypocenter, where the earthquake is felt with greatest intensity"
- **Seaquake:** earthquake with epicenter on the ocean floor, conducive to tsunami formation
- **Tsunami:** "large waves that travel at great speed, caused by an earthquake"

Earthquake propagation:

- **Seismic waves:** "earthquakes propagate through seismic waves, which lose energy as they move away from the hypocenter"

Device that detects earthquake occurrence:

- **Seismograph:** detects seismic waves and produces a graph – the seismogram

Scales used to measure an earthquake:

Richter Scale:

- Measures magnitude (released energy)
- Has no limit
- Based on study of seismograms

European Macroseismic Scale:

- Measures intensity (effects)
- Scale from I to XII
- Based on study of earthquake consequences
- Has subjective character
- Intensity values are recorded on isoseismal maps (lines connecting points of equal intensity)

Effects of an earthquake:

- Road cracks
- Land displacement
- Infrastructure destruction
- Injuries and deaths

Seismic activity and Earth's internal dynamics:

Main seismic zones: - Pacific Ring of Fire - Mid-Atlantic Ridge - Mediterranean-Asian Belt

Seismic activity is a consequence of Earth's internal dynamics, as is volcanism.

2.3.5.2 Earthquakes in Portugal

Seismic risk:

- Due to proximity to tectonic plate boundaries (African, Eurasian, and North American)
- Generally increases from north to south in mainland Portugal

Regions with highest seismic risk in Portugal: - Azores - Lisbon and Tagus Valley - Alentejo Coast - Algarve Coast

Seismic risk maps identify areas of greatest risk.

2.3.5.3 Earthquake prediction and population protection

Science and technology and earthquakes:

- "Earthquake prediction is not yet possible to predict an earthquake, but science and technology have contributed to reducing mortality"
- Anti-seismic construction
- Early earthquake detection and rapid alert to population at risk

Prevention measures:

Before an earthquake occurs: - Know where to turn off water, gas, and electricity - Always have a radio (battery-powered), flashlight, and first-aid kit at hand - Have reserves of water and canned and packaged food - Place heavier objects on lower shelves

During an earthquake: - Go under a table, bed, or doorway - Kneel and cover head with hands - Stay calm

After an earthquake: - Turn off water, gas, and electricity - Do not light matches, lighters, or candles - Do not use elevators - Listen to radio and follow authorities' recommendations

Essential Learning:

- Distinguish hypocenter from seismic epicenter and intensity from seismic magnitude
- Distinguish Richter Scale from European Macroseismic Scale
- Interpret seismograms and national isoseismal maps
- Discuss measures for protecting property and people (before, during, and after an earthquake)
- Explain the importance of science and technology in seismic prediction
- Explain the distribution of earthquakes and volcanoes on planet Earth

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2.4 Earth's History

2.4.1 Fossils and rock dating

2.4.1.1 Fossils

Paleontology and Fossils:

- **Paleontology**: science that studies fossils
- **Fossils**: remains of past living organisms, or traces of their activity

Fossil Classification:

- **Body fossils**: organism remains (bones, teeth, shells, trunks, leaves)
- **Trace fossils**: evidence of activity (eggs, excrement, footprints, nests, tunnels, seeds)

Importance of Fossils: Fossils allow understanding evolution of life and reconstructing Earth's history.

Fossilization Process: Fundamental conditions include: - Presence of skeleton (internal or external) - Rapid burial with fine sediments - Low temperature - Lack of oxygen - Absence of decomposers and scavengers

Fossilization Methods:

- **Mineralization**: replacement of organic matter with minerals
- **Molding**: organism's shape impressed in rocks (internal/external molds)
- **Conservation**: partial/total preservation in materials like resin or ice

Index Fossils vs. Facies Fossils:

- **Index fossils (or reference fossils)**: short period of existence, wide geographical distribution, used to date rocks (trilobites—Paleozoic; ammonites—Mesozoic)
- **Facies fossils (or environmental fossils)**: indicators of specific habitats, characterize paleoenvironments (corals—shallow and warm waters)

2.4.1.2 Rock dating

Absolute vs. Relative Dating:

- **Absolute (or radiometric) dating**: based on element transformation, dates rocks in thousands/millions of years
- **Relative dating**: comparison with other rocks using stratigraphic principles

Stratigraphic Principles:

- **Principle of Horizontality**: sediments deposit in horizontal layers
- **Principle of Superposition**: lower strata are older
- **Principle of Intrusion**: structures that intersect (faults, dikes) are more recent than affected strata

- **Principle of Paleontological Identity**: strata with the same types of fossils have the same age

Essential Learning: Students should identify fossilization stages, establish analogies with the real world, and explain the contribution of fossils to reconstructing Earth's history.

2.4.2 Stages of Earth's History

2.4.2.1 Earth's History

Historical time and geological time:

- Historical time: "refers to human history" and is measured in years, decades, centuries, or millennia
- Geological time: "refers to Earth's history" and is measured in millions of years

Geological time scale:

- Geochronological scale organizes Earth's history into eras divided into periods
- Beginning or end of each era relates to mass extinctions

Geological events that marked extinctions:

- Glaciations
- Marine transgressions
- Marine regressions
- Impact of cosmic bodies
- Intense volcanic activity

Earth's age: "about 4,600 million years"

Main stages:

Precambrian: Formation of lithosphere, atmosphere, and oceans; first unicellular marine life forms; diversity of multicellular invertebrates

Paleozoic Era: Large increase in marine biodiversity; conquest of terrestrial environment; emergence of non-flowering land plants, insects, fish, amphibians, and reptiles

Mesozoic Era: Development of reptiles and dominance of dinosaurs; emergence of flowering plants, mammals, and birds

Cenozoic Era: Development of large mammals; appearance of hominids

Essential Learning:

- Distinguish historical time from geological time in various documents
- Explain principles of geological reasoning and relative dating

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2.5 Geological Sustainability

2.5.1 Sustainable exploitation of rocks

2.5.1.1 Rocks in Portugal

Geological Map:

The geological map is a map that presents information regarding predominant rock materials and geological phenomena affecting them, in a given region.

Scientific Areas Contributing to Geological Map Preparation:

- Paleontology
- Geochemistry
- Petrography
- Topography
- Computer science

Distribution of Rock Types in Portugal:

- Plutonic igneous rocks: Northern and central regions
- Volcanic igneous rocks: Azores and Madeira
- Metamorphic rocks: North, center, and Alentejo
- Sedimentary rocks: Western and southern coastal strips, and Tagus and Sado basins

Applications of Rocks in Portugal:

- Construction: Slate, marble, granite, limestone
- Paving: Granite
- Glass manufacturing: Sand
- Ceramics: Clay
- Food industry: Rock salt
- Cosmetics: Pumice stone
- Energy production: Coal

2.5.1.2 Sustainability in Rock Extraction

Extraction Impacts:

Positive impacts include benefit to the country's economy.

Negative impacts involve landscape destruction, pollution, and concerns about resource depletion for future generations, since rocks represent non-renewable resources.

Requirements for Sustainable Exploitation:

- Reduce waste
- Recycle materials
- Search for alternative resources

- Develop pollution prevention technologies

2.5.2 The contribution of geological knowledge

2.5.2.1 Geological environment, health, and sustainability

Geological Environment:

The geological environment is defined as a location shaped by geological materials and processes, including rocks, soil, water, rock erosion, and soil formation.

Health Influences:

The presence or absence of substances in water, soil, and air affects the survival of living organisms and quality of life of populations. Examples include:

- Deficiency of nitrogen, potassium, and iron causing plant diseases
- Iodine deficiency impacting thyroid function and cognitive development
- Radon gas accumulation in buildings in uranium-rich regions
- Heavy metals (lead, cadmium, mercury) contaminating soils and waters

Impacts of Human Activity:

Consequences of Agriculture: Water pollution and soil salinization lead to habitat destruction, organism mortality, desertification, reduced soil productivity, and water scarcity.

Impacts of Mining and Quarrying: Soil and water contamination with heavy metals, land and watercourse alterations, inhalation of toxic dust, increased lung diseases, and water quality problems.

Effects of Population Growth: Increased resource consumption, waste production, pollution, and threats to sustainability.

Importance of Environmental Geology:

This field examines interactions between Earth's subsystems (geosphere, atmosphere, hydrosphere, biosphere) and human activities. Geological knowledge supports informed decisions about selecting sites for landfills, asbestos removal, mine remediation, and radioactivity assessment.

Essential learning: Relate geological environment to health and explain the importance of geological knowledge for Earth's sustainability.

Chapter 3

CELL BIOLOGY AND ECOLOGY

3.1 Conditions for Life

3.1.1 Earth conditions that allow the existence of life

3.1.1.1 LIFE ON EARTH

Earth's Position in the Solar System:

- Earth is the third planet closest to the Sun

Conditions that allow life on Earth:

- **appropriate distance from the Sun:** allows mild temperature and the existence of water in three physical states
- **appropriate mass:** allows retention of an atmosphere
  - important for respiration of living organisms
  - protects Earth against meteors
  - protects living organisms from ultraviolet rays (through the ozone layer)
  - has a moderating effect on temperature (due to greenhouse effect)
- **rocky substrate:** serves as support for life
- **magnetic field:** serves as shield against solar particles

3.1.1.2 EVOLUTION OF THE PLANET AND CREATION OF CONDITIONS FOR EXISTENCE OF LIFE

How did the planet evolve to allow life on Earth?

1. **Earth's formation:** intense volcanic activity and surface covered with lavas
2. **Cooling and solidification of Earth's surface and ocean formation:** water vapor existing in the atmosphere condensed and precipitated forming oceans where organic matter

formed

3. **Reduction of volcanic activity and appearance of first living organisms:** emergence of first living organisms, in oceans, from organic matter
4. **Evolution of life in oceans:** emergence of first photosynthetic organisms, which will alter atmospheric composition
5. **Formation of ozone layer and appearance of terrestrial living organisms:** atmosphere comes to have a composition similar to current, which allowed appearance of terrestrial living organisms
6. **Evolution of biodiversity:** life evolves and diversifies until present

3.1.1.3 THE IMPORTANCE OF THE ATMOSPHERE

How did Earth's atmosphere evolve since its formation?

primitive atmosphere:

- main constituents:
  - nitrogen (65%)
  - water vapor (25%)
  - carbon dioxide (5%)
  - methane and ammonia (5%)
- this atmospheric composition was mainly due to intense volcanic activity

current atmosphere:

- main constituents:
  - nitrogen (78%)
  - oxygen (21%)
  - water vapor (0.4%)
  - carbon dioxide (0.04%)
- amounts of methane and ammonia decreased until disappearing
- appearance of photosynthetic living organisms caused carbon dioxide amount to decrease and oxygen to appear in increasingly larger amounts in atmosphere

What is the importance of the atmosphere?

- **allows gas exchanges with living organisms essential for their survival:**
  - in respiration living organisms receive oxygen and release carbon dioxide
  - in photosynthesis plants receive carbon dioxide and release oxygen
- **protects Earth against meteors:** rocky bodies from Space, upon entering atmosphere, ignite and disintegrate
- **protects living organisms against ultraviolet radiation:** due to existence of ozone layer (O) that prevents passage of these rays harmful to living organisms

- **has a temperature moderating effect:** due to greenhouse effect, that is, retention of energy from the Sun by some atmospheric gases

One of the main greenhouse gases is carbon dioxide. This is why variation in planet temperature over time is related to concentration of this gas in atmosphere. The higher the carbon dioxide concentration, the higher the temperature recorded on the planet.

Curricular Goals:

Students should: 1. Identify Earth's position in the Solar System 2. Explain three Earth conditions that allowed development and maintenance of life 3. Interpret graphs of temperature evolution, solar energy, and atmospheric carbon dioxide 4. Describe influence of living organisms' activity on atmospheric evolution 5. Infer importance of greenhouse effect for maintaining temperature favorable to life

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3.2 Cell Organization

3.2.1 The microscope

Note: Content about "The microscope" is listed in the 8th grade curricular structure but does not have an active content page available on O Bichinho do Saber website.

This topic serves as methodological introduction before studying the cell and biological organization, preceding detailed study of cellular structures and levels of organization of living organisms.

3.2.2 The cell and levels of biological organization

3.2.2.1 Fundamental Concepts

Habitat: "place where a given living organism inhabits by finding necessary conditions"

Ecosystem: An ecosystem comprises: - **Biotic components:** living organisms and their relationships - **Abiotic components:** physical environment and factors such as water, light, temperature

3.2.2.2 Hierarchy of Biological Organization

Biological organization is structured from smallest to largest level:

cell → tissue → organ → organ system → organism species → population → community → ecosystem

Species: "set of generally similar organisms that, when crossed with each other, originate fertile offspring"

Population: "set of individuals of same species that inhabit a given place (habitat) and that interact with each other"

Community: "set of populations that inhabit a given place (habitat) and that interact with each other"

Ecosystem: "set of living organisms (community), environment where they inhabit (habitat), relationships with each other and with environment"

3.2.2.3 Biome Classification

Terrestrial Biomes:

- Tundra
- Taiga
- Temperate forest
- Tropical forest
- Savanna
- Chaparral

Aquatic Biomes:

- Oceans
- Mangroves
- Swamps
- Rivers

Biomes are "sets of interacting ecosystems that are conditioned by geographical conditions and region's climate."

3.3 Ecosystems

3.3.1 Levels of ecosystem organization

3.3.1.1 Ecosystem Components

The ecosystem contains two essential elements:

- **Biotic component**: living organisms and their interactions
- **Abiotic component**: physical environment including water, light, temperature, soil, and wind

Curricular Objective: Understand levels of biological organization of ecosystems, presenting definitions and describing structure, functioning, and equilibrium.

3.3.2 Abiotic factors (environmental influence on living organisms)

"Abiotic factors are environmental factors that influence living organisms."

The five main abiotic factors include light, temperature, water, soil, and wind.

3.3.2.1 INFLUENCE OF LIGHT

Key concepts:

- **Phototropism**: "slow movement of plants toward sunlight"
- **Heliophilous plants**: develop under direct and intense light
- **Shade plants**: need shade (examples: ferns and mosses)

Classification by photoperiod:

- Long-day plants (flowering > 12h light)
- Short-day plants (flowering < 8h light)
- Indifferent plants (not influenced)

Vertical stratification:

- Herbaceous layer (lower layer)
- Shrub layer (middle layer)
- Tree layer (upper layer)

Influence on animals:

- **Photophilic animals**: attracted by light (positive phototaxis) - examples: moths, butterflies
- **Photophobic animals**: avoid light (negative phototaxis) - examples: bat, earthworm
- **Diurnal animals**: active during day
- **Nocturnal animals**: active at night

Migrations: "Regular movements of a living organism to places that provide better survival conditions"

3.3.2.2 INFLUENCE OF TEMPERATURE

Thermal amplitude and tolerance:

- **Tolerance range**: temperature limits for survival
- **Optimum temperature**: ideal development conditions
- **Lethal temperature**: leads to organism death

Classification of living organisms:

- **Eurythermic organisms**: large thermal amplitude (example: wolf)
- **Stenothermic organisms**: small thermal amplitude (example: snake)

Homeothermy vs. Poikilothermy:

- **Homeothermic organisms**: maintain constant body temperature (mammals, birds)
- **Poikilothermic organisms**: body temperature varies with environment (fish, reptiles, amphibians)

Behavioral adaptations:

- **Estivation**: "reduction of activity during hot season" (snail, crocodile)
- **Hibernation**: "reduction of activity during cold season" (hedgehog, bear)
- **Migration**: regular movement to better conditions

Body adaptations:

- Hot environments: large ears, short fur
- Cold environments: small ears, long fur, thick fat layer

Physiological adaptations:

- Hot: panting to lose heat
- Cold: hair erection for thermal insulation

Adaptations in plants:

- Deciduous plants (ex: chestnut tree)
- Reduction to underground part (ex: narcissus)
- Reduction to seeds (ex: corn)

3.3.2.3 INFLUENCE OF WATER

Classification of living organisms:

- **Hydrophilic organisms**: live permanently in water (octopus, water lily)
- **Hygrophilic organisms**: live in humid places (earthworm, rice paddy)
- **Mesophilic organisms**: need moderate amounts (horse, oak tree)
- **Xerophilic organisms**: live in dry places (dromedary, cactus)

Adaptations in plants to dry climate:

- Long and shallow roots
- Fleshy stems to accumulate water
- Small leaves or reduced to spines

Adaptations in animals:

- Fat reserves to produce water (dromedary, camel)
- Waterproof covering (scorpion)
- Do not sweat, little urine, nocturnal activity (gerbil)

3.3.2.4 INFLUENCE OF SOIL

Definition: "most superficial layer of Earth's crust, consisting of organic matter, mineral matter, water, and air"

Aquatic substrates:

- Soft (sands in rivers and oceans)
- Hard (rocks with limpets and mussels)

Factors influencing distribution:

- Porosity
- Composition
- Permeability
- Humidity
- Texture

Importance: Functions as habitat and fixation medium for plants, water capture, and essential minerals.

3.3.2.5 INFLUENCE OF WIND

Main functions:

- Seed dispersal
- Transport of bacteria, fungi, and spores
- Aeration of aquatic waters
- Ocean wave formation

Influence on morphology: Windy regions present creeping plants and small, flattened animals.

Behavioral influence:

- Grasshoppers in clouds take advantage of air displacement
- Birds of prey glide in hot air masses
- Migration routes influenced by prevailing winds

Curricular Goals: 1. Describe influence of five abiotic factors 2. Present examples of adaptations to factors 3. Test variables in laboratory 4. Conclude about variable influence 5. Predict influences in local ecosystems 6. Relate environmental changes with evolution or extinction of species

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3.3.3 Biotic factors (relationships between living organisms)

3.3.3.1 WHAT ARE BIOTIC RELATIONSHIPS

Biotic relationships:

- Interactions between living organisms that are part of an ecosystem

Intraspecific interactions:

- Interactions between living organisms of same species

Interspecific interactions:

- Interactions between living organisms of different species

3.3.3.2 INTRASPECIFIC INTERACTIONS

Cooperation (+,+)

- Individuals contribute to common benefit of group
- *Ex: Meerkats organize themselves in vigilance and food collection*

This type of relationship is established when organisms of same population organize themselves into **societies** *(ex: bees)* or **colonies** *(ex: penguins)*. In a society there is hierarchical organization and division of tasks, while in a colony there is no hierarchy or differentiated functions.

Competition (-,-)

- Harmful to living organisms involved
- *Ex: Hippos compete for attention of females*

Reasons: struggle for water, light, food, territory, and female.

The maximum expression of intraspecific competition is **cannibalism**, in which one individual kills another of same species, feeding on it.

3.3.3.3 INTERSPECIFIC RELATIONSHIPS

Facultative mutualism (+,+)

- Benefit for both species, although relationship is not fundamental for survival
- *Ex: Cleaner wrasse cleans moray eel taking advantage of its food remains*

Obligatory mutualism (+,+)

- Benefit for both species, being this a fundamental relationship for survival
- *Ex: association between unicellular algae and fungi (which originate lichens)*

Competition (-,-)

- Harmful to living organisms involved
- *Ex: Lions and hyenas compete for territory, water, and food*

Predation (+,-)

- Benefit for **predator** and harm for **prey**
- *Ex: Bear preys on salmon*

Characteristics favoring hunting: developed claws and strong curved beaks (ex: eagle) and sharp teeth and strong jaws *(ex: lion)*.

Both predators and prey may also have **camouflage** or **mimicry** ability to go unnoticed. In camouflage they acquire aspects similar to nature *(ex: stick insect)*, and in mimicry they acquire aspects similar to other living organisms *(ex: false coral snake)*.

Parasitism (+,-)

- Benefit for **parasite** and harm for **host**
- *Ex: A tapeworm can be found in intestine of various animals*

Parasites are classified according to way of life: - **ectoparasites**, if they live outside host's body *(ex: flea)* - **endoparasites**, if they live inside host's body *(ex: tapeworm)*

They are also classified according to size: - **microparasite**, if they are small and with very short gestation times *(ex: viruses and bacteria)* - **macroparasites**, if they are larger and with longer gestation times *(fleas, ticks, tapeworms, and roundworms)*

Commensalism (+,0)

- Benefit for **commensal** and indifferent for other individual
- *Ex: Remora is transported by shark, takes advantage of its food remains*

Amensalism (-,0)

- Harm for **amensal** and indifferent for **inhibitor**
- *Ex: Penicillium fungus produces substances that inhibit bacteria development*

3.3.3.4 HOW CAN BIOTIC RELATIONSHIPS CONDITION ECOSYSTEM DYNAMICS

The number of individuals in a population in an ecosystem varies over time until reaching equilibrium. This equilibrium depends on relationships established between number of individuals, available space, amount of available food, and number of predators.

For example, if number of predators in a given ecosystem decreases, this will not mean it will be good for population of their prey. This is because number of individuals in that population will increase and there may no longer be food for so many individuals.

Learning Objectives: 1. Distinguish, giving examples, intraspecific from interspecific interactions 2. Identify types of biotic relationships, in various documents 3. Interpret graphs showing population dynamics resulting from biotic relationships 4. Evaluate consequences of some biotic relationships on ecosystem dynamics 5. Explain how biotic relationships can lead to evolution or extinction of species

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3.3.4 Food chains and matter cycles

3.3.4.1 Learning Objectives

According to the 8th grade Natural Sciences curriculum, students should:

- Explain how activities of living organisms — feeding, respiration, photosynthesis — affect matter cycles
- Understand importance of matter recycling in ecosystem dynamics
- Interpret main phases of water, carbon, oxygen, and nitrogen cycles using diagrams
- Justify how human activities interfere with matter cycles and impact ecosystems

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3.3.5 Ecological successions

3.3.5.1 DYNAMIC EQUILIBRIUM OF ECOSYSTEMS AND PLANET EARTH SUSTAINABILITY

Ecological succession: "sequence of communities that replace each other, over time in a given area."

Primary ecological succession: "emerges in a place where there was never life before"

Secondary ecological succession: "emerges in a place where there was life before"

Pioneer communities:

- lichens
- mosses
- grasses
- insects
- earthworms
- rodents

3.3.5.2 Curricular Objectives

Main Competency: Relate dynamic equilibrium of ecosystems with planet Earth sustainability

Specific Goals to Achieve:

1. Describe phases of ecological succession, using a concrete example
2. Distinguish primary from secondary ecological succession
3. Identify type of ecological succession described in various documents
4. Explain causes and consequences of alteration of dynamic equilibrium of ecosystems
5. Conclude about importance of dynamic equilibrium of ecosystems for life sustainability on planet Earth

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Chapter 4

Natural Sciences Manual: Grades 7 to 9

Thematic Organization

This manual consolidates the Natural Sciences content for grades 7, 8, and 9, organized by scientific domains rather than by school year.

Content adapted from: O Bichinho do Saber (www.obichinhodosaber.com)

4.1 SUSTAINABILITY AND ENVIRONMENTAL MANAGEMENT

4.1.1 Sustainable Development

- Sustainable development
- Natural resources

4.1.2 Environmental Protection

- Catastrophes
- Measures for ecosystem protection
- Nature protection and conservation

4.1.3 Territory and Resource Management

- Land use planning and management
- Waste and water management

4.2 HUMAN BIOLOGY

4.2.1 Health and Body Organization

- Individual and community health
- Structural levels of the human body

4.2.2 Nutrition and Digestion

- Foods, nutrients, and healthy eating
- Digestive system

4.2.3 Circulatory and Immune System

- Blood
- Cardiovascular system
- Lymphatic system

4.2.4 Respiratory System

- Respiratory system
- Basic life support

4.2.5 Excretory System

- Excretory system: urinary system
- Excretory system: skin

4.2.6 Coordination and Regulation

- Neuro-hormonal system: nervous system
- Neuro-hormonal system: hormonal system

4.2.7 Reproduction and Heredity

- Reproductive system
- Heredity

Chapter 5

SUSTAINABILITY AND ENVIRONMENTAL MANAGEMENT

5.1 Sustainable Development

5.1.1 Sustainable Development

Sustainable development is defined as: "development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

5.1.1.1 Ecosystem Services

Ecosystems provide various essential services for human life and for planet equilibrium. These services can be organized into four main categories:

Production Services:

- Products and consumer goods obtained directly from ecosystems
- Include: food, water, fuels, medicines
- Products used daily by population

Regulation Services:

- Natural processes that mitigate environmental risks
- Include: pollination, climate regulation, water purification, disease control
- Allow ecosystems to maintain their natural equilibrium

Cultural Services:

- Non-material benefits offered by ecosystems
- Include: tourism, leisure, cultural heritage, scientific research, education

- Contribute to psychological and social well-being of populations

Support Services:

- Fundamental processes that allow existence of all other services
- Include: soil formation, nutrient cycles, habitat provision
- Essential basis for life on planet

5.1.1.2 Components of Human Well-being

Adequate ecosystem management contributes to four main elements of human well-being:

1. **Security** - protection against environmental and natural risks
2. **Basic Material Resources** - access to food, water, and energy
3. **Health** - environmental conditions that promote physical and mental health
4. **Social and Cultural Relations** - opportunities for social interaction and cultural expression

5.1.1.3 Learning Objectives

According to official 8th grade Natural Sciences program, students should:

- Analyze importance of ecosystem management
- Understand concept of sustainable development
- Relate ecosystem management to human well-being
- Identify different services that ecosystems provide
- Recognize interdependence between human beings and ecosystems

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5.1.2 Natural resources

Definition of Natural Resources:

"goods found in nature that humans can use for their benefit"

5.1.2.1 Classification by Renewability

Renewable Resources:

"resources that are not at risk of being depleted on human life scale"

Examples: - Sunlight - Wind - Plants - Animals

Non-Renewable Resources:

"resources that are exploited at a rate exceeding their natural renewal rate, and may become depleted on a human timescale"

Examples: - Coal - Petroleum - Natural gas - Uranium - Minerals and rocks

5.1.2.2 Classification by Type

Mineral Resources:

- Metallic: wolframite, chalcopyrite, bauxite, magnetite
- Non-Metallic: ornamental rocks, industrial rocks, industrial minerals

Water Resources:

- Spring water
- Natural mineral water
- Thermal water
- Carbonated water

Biological Resources:

- Agricultural: animals and plants
- Forest: forest products
- Marine: algae, marine plants and animals

Energy Resources:

- Fossil fuels (petroleum, coal, natural gas)
- Uranium
- Wind
- Water

5.2 Environmental Protection

5.2.1 Catastrophes

Catastrophes can be classified into two main types:

5.2.1.1 Natural Catastrophes

Geological:

- Earthquakes
- Volcanism
- Ground movements

Climatic:

- Storms
- Floods
- Droughts
- Fires

5.2.1.2 Anthropogenic Catastrophes

Events caused by humans, including: - Deforestation - Pollution - Biological invasions - Transportation of hazardous materials

5.2.1.3 Specific Natural Phenomena

Seismicity: "Sudden movements" characterize earthquakes, while tsunamis form when seismic activity occurs in the ocean. Prevention measures include earthquake-resistant construction and public awareness.

Volcanism: Eruptions release lava causing direct habitat destruction, plus ash and gases that create pollution and infrastructure damage.

Ground Movements: These include rockfalls, landslides and fluid flows, with prevention methods involving drainage systems, planting vegetation and concrete stabilization.

Climatic Events: Storms, floods and droughts, each with specific consequences—floods cause water overflow and erosion, while droughts cause soil degradation and desertification.

5.2.1.4 Anthropogenic Issues

Deforestation: The removal of forests increases soil erosion, disperses pollutants and reduces biodiversity.

Invasive Species: Non-native species that cause significant changes in ecosystems can eliminate indigenous populations.

Air Pollution: Greenhouse effects, ozone layer depletion and acid rain—each with documented environmental consequences.

Water Pollution: Bioaccumulation moves pesticides and heavy metals through food chains at increasing concentrations, while eutrophication causes excessive nutrient accumulation that disrupts aquatic ecosystems.

5.2.1.5 Educational Objectives

Students should: - Distinguish between natural and anthropogenic catastrophes - Describe the causes of human-induced disasters - Understand how pollution and invasions affect ecosystem diversity and species extinction

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5.2.2 Ecosystem Protection Measures

5.2.2.1 Learning Objectives

Students should be able to:

1. **Synthesize ecosystem protection measures** and indicate three measures aimed at reducing the impacts of catastrophes of natural and anthropogenic origin on living beings and the environment
2. Categorize information about existing natural risks in the school region and anthropogenic occupation of that region
3. Identify protection measures for living beings and the environment in ecosystems near their region
4. Construct documents in various formats about protection measures implemented in their region and nearby ecosystems
5. Explain how each citizen can contribute to the effectiveness of protection measures

5.2.2.2 Public Conservation Organizations

Institute for Nature Conservation and Forests (ICNF):

The ICNF is the public organization responsible for managing the Natura 2000 Network in Portugal. The Natura 2000 Network is a European initiative that seeks to "ensure the long-term conservation of the most threatened species and habitats."

5.2.2.3 Associations and Non-Governmental Organizations (NGOs)

The following organizations work in environmental protection in Portugal:

- **League for the Protection of Nature (LPN)**
- **FAPAS** (Fund for the Protection of Wild Animals)
- **Quercus** (National Association for Nature Conservation)
- **Portuguese Society for the Study of Birds (SPEA)**
- **GEOTA** (Group for Territorial Planning and Environmental Studies)

5.2.2.4 Important International Treaties

The main international conventions and protocols for environmental protection include:

Treaty/Conference	Year	Main Objectives
Stockholm Conference	1972	Recommendations against atmospheric and aquatic pollution
Montreal Protocol	1985	Reduction of CFCs and ozone layer protection
ECO 92 Conference (Rio)	1992	Inventories of threatened species; biodiversity

Treaty/Conference	Year	Main Objectives
Kyoto Protocol	1997	Reduction of greenhouse gas emissions
Rio + 10 (Johannesburg)	2002	Reinforcement of sustainable development
COP Conferences	2009-2013	Continuation of environmental protection goals

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5.2.3 Nature Protection and Conservation

5.2.3.1 Public Organizations for Nature Protection and Conservation

Institute for Nature Conservation and Forests (ICNF):

The Institute for Nature Conservation and Forests (ICNF) is the organization responsible for managing "the ecological network for the space of the European Union that aims to ensure the long-term conservation of the most threatened species and habitats in Europe" through the Natura 2000 Network.

5.2.3.2 Portuguese Nature Protection Associations

The main Environmental Non-Governmental Organizations (ENGOs) in Portugal include:

- **League for the Protection of Nature (LPN)**
- **FAPAS** (Fund for the Protection of Wild Animals)
- **Quercus**
- **Portuguese Society for the Study of Birds (SPEA)**
- **GEOTA** (Group for Territorial Planning and Environmental Studies)

5.2.3.3 International Environmental Treaties and Agreements

Nature protection is regulated by various international treaties and conferences:

- **Stockholm Conference (1972)** - Initial milestone of global environmental conferences
- **Montreal Protocol (1985)** - Focused on CFC reduction and ozone layer protection
- **Earth Summit - ECO 92 (Rio de Janeiro)** - Agenda 21 and sustainable development
- **Kyoto Protocol (1997)** - Commitments on greenhouse gas emission reductions
- **Rio +10 (Johannesburg, 2002)** - Assessment of progress in Agenda 21 implementation
- **Rio +20 (Rio de Janeiro, 2012)** - Sustainable development and green economy
- **COP Conferences (Conference of the Parties)** - Subsequent meetings of signatory parties to environmental treaties

5.2.3.4 Curricular Objectives

According to the curricular goals of the "Sustainability on Earth" domain, students should be able to:

- List Portuguese public organizations and agencies for nature protection and conservation
- Construct syntheses on regional environmental problems and mitigation strategies
- Understand the role of international treaties and agreements in environmental protection

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5.3 Territory and Resource Management

5.3.1 Land Use Planning and Management

5.3.1.1 Public Conservation Organizations

The Portuguese government established the **Institute for Nature Conservation and Forests (ICNF)** to manage conservation efforts. This institute oversees "the set of locations in Portugal that are part of the Natura 2000 Network."

The **Natura 2000 Network** functions as an ecological network throughout the European Union, designed to "ensure the long-term conservation of the most threatened species and habitats in Europe."

5.3.1.2 Non-Governmental Associations (ENGOs)

Portugal hosts several environmental protection associations: - League for the Protection of Nature (LPN) - FAPAS - Quercus - Portuguese Society for the Study of Birds (SPEA) - GEOTA

5.3.1.3 International Environmental Agreements

The main documented treaties include:

- **Stockholm (1972)**: Recommendations against air and water pollution and nuclear testing
- **Montreal Protocol (1985)**: CFC reduction measures
- **Rio 92 (1992)**: Inventories of endangered species
- **Kyoto Protocol (1997)**: Reduction of greenhouse gas emissions
- **Rio +10 (2002)**: Reinforcement of sustainable development commitment
- **Rio +20 (2012)**: Continuation of sustainability goals
- **COP 15 (2009)**: Climate Change Conference
- **COP 18-19 (2012-2013)**: Climate Change Conferences

5.3.2 Waste and Water Management

5.3.2.1 Learning Objectives

Students should master the following competencies:

1. **Distinguish the various types of waste** - Identify and classify different categories of waste generated by human activities
2. **Understand the importance of collection, treatment and sustainable waste management** - Summarize the importance of promoting collection, treatment and sustainable waste management
3. **Plan awareness campaigns** - Plan awareness campaigns on sustainable waste management
4. **Construct action plan for reducing water consumption** - Construct an action plan aimed at reducing water consumption at school and at home, based on the European Water Charter
5. **Propose measures to reduce water contamination** - Propose measures to reduce risks of water contamination resulting from human activities

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Chapter 6

HUMAN BIOLOGY

6.1 Health and Body Organization

6.1.1 Individual and Community Health

6.1.1.1 Health and Quality of Life

World Health Organization (WHO):

The **World Health Organization (WHO)** is a United Nations agency founded on April 7, 1948, dedicated to promoting global health.

The WHO defines health as "**a state of physical, mental and social well-being and not merely the absence of disease or infirmity**".

Quality of Life:

Quality of life is the personal perception of one's own position in life, considering cultural context, value system and personal expectations.

It is a concept that is: - **Subjective** - varies from person to person - **Dynamic** - changes over time - **Multidimensional** - encompasses multiple aspects of life

Quality of life encompasses six main domains:

1. **Physical** - health, energy, pain, discomfort
2. **Psychological** - self-esteem, body image, negative and positive feelings
3. **Level of dependence** - ability to live with autonomy
4. **Social relationships** - personal relationships, social support, sexual activity
5. **Environment** - safety, environmental quality, financial resources
6. **Personal beliefs** - religiosity, spirituality, values and meaning of life

6.1.1.2 Pathogenic Agents

Disease-causing agents (pathogenic agents) are classified into three main categories:

Physical Agents:

- Excessive noise
- Radiation (ultraviolet rays, X-rays, ionizing radiation)
- Weight lifting
- Physical trauma

Chemical Agents:

- Pesticides
- Dioxins
- Bisphenol A (BPA)
- Nicotine (tobacco)
- Asbestos
- Industrial chemicals

Biological Agents:

- **Bacteria** (e.g., *Salmonella*, *Mycobacterium tuberculosis*)
- **Viruses** (e.g., influenza, COVID-19, hepatitis)
- **Protozoa** (e.g., *Plasmodium* - malaria, *Giardia*)
- **Fungi** (e.g., *Candida*, *Aspergillus*)
- **Worms** (e.g., tapeworms, roundworms)

6.1.1.3 Communicable Diseases

Communicable diseases are those caused by infectious agents that spread from one person to another.

Transmission Routes:

- **Fecal-oral route** - ingestion of contaminated food or water
- **Airborne route** - inhalation of respiratory droplets
- **Direct route** - direct contact with skin or mucous membranes
- **Bloodborne route** - transfusion of contaminated blood, sharing needles
- **Vectors** - insects such as mosquitoes, ticks or lice

Examples of Communicable Diseases:

- Tuberculosis
- Influenza
- Measles
- Hepatitis B
- AIDS/HIV
- Malaria
- Dengue
- Cholera
- Gastrointestinal infections (gastroenteritis)

Responsible Use of Antibiotics:

Antibiotics are medications that fight bacterial infections. It is essential to use them responsibly:

- **Should only be used with a medical prescription**
- **Respect recommended dosages**
- **Complete the full treatment period**, even if you feel better
- **Never share antibiotics** with other people

Why? Improper use of antibiotics leads to the development of **bacterial resistance**, making bacteria immune to medications and making it difficult to treat future infections.

6.1.1.4 Non-Communicable Diseases

Non-communicable diseases are not caused by infectious agents and are not transmitted between people.

Main Non-Communicable Diseases:

1. **Cardiovascular Diseases**
   - High blood pressure
   - Myocardial infarction
   - Stroke
   - Cardiac arrhythmias
2. **Cancer**
   - Lung cancer
   - Breast cancer
   - Colon cancer
   - Skin cancer
3. **Diabetes**
   - Type 1 diabetes
   - Type 2 diabetes
4. **Chronic Respiratory Diseases**
   - Asthma
   - Chronic Obstructive Pulmonary Disease (COPD)
   - Emphysema
5. **Digestive System Diseases**
   - Peptic ulcers
   - Crohn's disease
   - Irritable bowel syndrome

Risk Factors:

The main non-communicable diseases are associated with:

- **Smoking** - smoking is one of the biggest risk factors
- **Sedentary lifestyle** - lack of regular physical activity

- **Poor diet** - excessive consumption of fats, sugars and salt
- **Alcoholism** - excessive consumption of alcoholic beverages
- **Obesity** - excess body weight
- **Chronic stress** - prolonged and unmanaged stress
- **Genetic factors** - hereditary predisposition
- **Aging** - advanced age

6.1.1.5 Health Promotion

Health promotion involves creating conditions for people to achieve better health status.

Health Determinants:

Health determinants are factors that influence individual and community health:

Biological Determinants:

- Genetic makeup
- Age
- Sex
- Immune status

Behavioral Determinants:

- Eating habits
- Physical activity levels
- Tobacco and alcohol consumption
- Treatment adherence

Environmental Determinants:

- Water and air quality
- Basic sanitation
- Environmental safety
- Exposure to pollutants

Socioeconomic Determinants:

- Economic income
- Education level
- Employment
- Access to health care
- Housing conditions
- Social inclusion

Health Promotion Measures:

1. Awareness Campaigns:

- Health education
- Information dissemination about diseases
- Promotion of healthy lifestyles

2. Vaccination:

- Systematic vaccination programs
- Vaccination campaigns against specific diseases
- Individual and community protection

3. Screenings:

- Cancer screenings (breast, colon, cervix)
- Blood pressure assessment
- Blood tests and disease screening

4. Basic Sanitation:

- Supply of drinking water
- Wastewater treatment
- Personal and sanitary hygiene
- Vector control

5. Environmental Quality:

- Reduction of air pollution
- Protection of natural resources
- Green spaces and leisure
- Food safety

6. Promotion of Healthy Habits:

- Balanced and varied diet
- Regular physical activity (at least 150 minutes per week)
- Reduction of salt, sugar and saturated fat consumption
- Smoking cessation
- Reduction of alcohol consumption
- Stress management
- Sleep 7-9 hours per night
- Personal hygiene

7. Access to Health Care:

- Regular medical consultations
- Affordable medications
- Quality treatments
- Mental health support

Healthy Lifestyles:

A healthy lifestyle should include:

- **Adequate nutrition** - consumption of fruits, vegetables, whole grains, lean proteins
- **Regular exercise** - cardiovascular activities, strength and flexibility
- **Sufficient rest** - regular, quality sleep
- **Stress management** - relaxation techniques, meditation, leisure activities
- **Social relationships** - maintaining meaningful connections
- **Prevention of risk behaviors** - avoiding tobacco, alcohol and drugs
- **Personal hygiene** - hand washing, dental hygiene, body hygiene
- **Safety** - safe practices at home, at work and in the community

6.1.1.6 Summary of Key Concepts

1. **Health** is a state of physical, mental and social well-being, not just absence of disease
2. **Quality of life** is subjective and involves multiple dimensions of life
3. **Pathogenic agents** (physical, chemical, biological) cause diseases
4. **Communicable diseases** spread from person to person
5. **Non-communicable diseases** are related to lifestyles and genetic factors
6. **Health promotion** involves education, prevention and access to care
7. **Healthy lifestyles** are fundamental to prevent diseases

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6.1.2 Structural Levels of the Human Body

Note: The content on "Structural levels of the human body" is listed in the 9th grade curriculum structure but does not have an active content page available on the O Bichinho do Saber website.

This topic serves as a bridge between the study of the cell (from 8th grade) and specific organ systems (from 9th grade), presenting the organizational hierarchy:

cell → tissue → organ → organ system → organism

6.2 Nutrition and Digestion

6.2.1 Foods, Nutrients and Healthy Eating

6.2.1.1 Foods and Nutrients

Fundamental Definitions:

Foods are "natural or processed products, solid or liquid, used by living beings as a source of matter and energy."

Nutrients are "the molecules that make up foods, essential for the functioning of the human body."

Classification of Nutrients:

According to size, nutrients are divided into macromolecules and micromolecules. There are six main groups:

1. **Proteins** - composed of proteins, peptides and amino acids. Sources: meat, fish, legumes, eggs, milk
2. **Carbohydrates** - simple and complex sugars (starch, fibers, glucose). Sources: bread, potatoes, rice, pasta, fruits
3. **Lipids** - triglycerides composed of fatty acids and glycerol. Sources: fatty fish, nuts, butter, olive oil
4. **Water** - essential micromolecules
5. **Mineral salts** - micromolecules, abundant in fruits and vegetables
6. **Vitamins** - regulatory micromolecules

Identification Tests:

- Proteins: react with "sodium hydroxide and copper sulfate, giving rise to a violet solution"
- Starch: turns "lugol solution or iodine water dark blue"
- Glucose: reacts with "fehling liquor, giving rise to a brick-colored precipitate"
- Lipids: react with "sudan III solution, giving rise to reddish-colored fat droplets"
- Vitamin C: reacts with "indophenol, which becomes colorless"

Organic/Inorganic Classification:

Organic nutrients (proteins, carbohydrates, lipids, vitamins) are produced by living beings and contain carbon. Inorganic nutrients (water, mineral salts) are found in nature without carbon.

Macronutrients vs. Micronutrients:

Macronutrients exist in greater quantities (proteins, lipids, carbohydrates, water). Micronutrients exist in smaller quantities (vitamins, mineral salts).

6.2.1.2 Functions of Nutrients

Energy Function: Carbohydrates provide main energy through cellular respiration. Lipids produce caloric energy.

Plastic Function: "Proteins, lipids, water and mineral salts are the main constituents of cells."

Regulatory Function: "Water, mineral salts, vitamins, proteins and fibers" facilitate essential chemical reactions.

6.2.1.3 Mineral Salts and Trace Elements

Macrominerals: Potassium, chlorine, calcium (needed in larger quantities)

Microminerals (trace elements): Iron, fluorine, iodine (needed in smaller quantities)

Deficiencies and Consequences:

1. **Iron** - participates in "oxygen and carbon dioxide transport". Deficiency causes anemia (fatigue, sleep disturbances, hair loss)
2. **Fluorine** - is part of "bones and teeth". Deficiency causes dental cavities
3. **Iodine** - is part of "hormones and the thyroid". Deficiency causes goiter and developmental delays

6.2.1.4 Eating Disorders

Eating disorders are "diseases of a psychic nature that cause changes in an individual's food behavior patterns."

Types:

- **Binge eating** - uncontrolled ingestion leading to obesity and cardiovascular diseases
- **Anorexia nervosa** - abnormal body perception, strict dieting
- **Bulimia nervosa** - excessive ingestion followed by vomiting and use of laxatives

6.2.1.5 Healthy Eating

The **Mediterranean diet** is a "healthy and balanced dietary regimen, typical of peoples of Mediterranean culture." The pyramid indicates foods to consume in different quantities, considering seasonal products and emphasizing physical activity and socializing.

6.2.2 Digestive System

6.2.2.1 Constitution of the Digestive System

The digestive system comprises:

Digestive Tract:

- Mouth
- Pharynx
- Esophagus
- Stomach
- Small intestine
- Large intestine
- Rectum
- Anus

Accessory Glands:

- Salivary glands
- Liver (with gallbladder)
- Pancreas

6.2.2.2 Nutrition

Definition: "Process through which the organism assimilates nutrients existing in foods, so that it can use them in its metabolism"

Stages of Nutrition:

Nutrition occurs in three stages:

1. **Ingestion** - introduction of food
2. **Digestion** - transformation of complex molecules into simpler substances
3. **Absorption** - passage of nutrients into blood and lymph

6.2.2.3 Digestion Process

Digestion involves physical transformations through mechanical actions and chemical transformations by enzyme action. It occurs in multiple locations of the digestive tract.

Mouth:

Physical Actions:

- Mastication

Chemical Actions:

- Salivation
- Salivary amylase transforms starch into maltose

Result: Food bolus

Stomach:

Physical Actions:

- Peristaltic movements

Chemical Actions:

- Gastric juice
- Pepsin transforms proteins into peptides
- Lipases transform lipids

Result: Chyme

Duodenum:

Chemical Agents:

- Pancreatic juice
- Intestinal juice
- Bile

Enzymes and Their Actions:

- Pancreatic amylase
- Trypsin
- Peptidase
- Lipases
- Maltase
- Lactase
- Sucrase

Result: Chyle

6.2.2.4 Intestinal Absorption

Nutrient absorption occurs mainly through the walls of the small intestine, which contains **valvulae conniventes** with **intestinal villi** and **microvilli**, which increase the absorption surface.

Absorption by Blood Capillaries:

- Water
- Water-soluble vitamins
- Glucose
- Minerals
- Glycerol
- Amino acids

Absorption by Lymphatic Vessels:

- Fatty acids
- Fat-soluble vitamins

6.2.2.5 Intestinal Microbiota

Beneficial microorganisms present in the intestine: - Synthesize amino acids and vitamins - Decompose undigested nutrients - Protect against pathogenic microorganisms

6.2.2.6 Diseases of the Digestive System

Peptic Ulcer: Lesion caused by: - Excess acid in the stomach - Decreased protection of the wall

Gastritis and Gastroenteritis: Inflammations caused by: - Viral infections - Contaminated foods

Cancer: Tumors that can occur in: - Stomach - Colon - Rectum

Risk Factors:

- Genetic factors
- Alcohol
- Tobacco
- Diets low in fiber

6.2.2.7 Recommendations for Digestive Health

- Diet rich in fiber, vegetables and fruit
- Proper mastication
- Tooth brushing after meals

Learning Objectives:

1. Identify the stages of nutrition
2. Relate digestive function to cellular metabolism
3. Describe physical and chemical transformations in digestion
4. Justify the role of valvulae conniventes
5. Identify three diseases of the digestive system
6. Propose functional measures for digestive health
7. Explain the importance of intestinal microbiota

6.3 Circulatory and Immune System

6.3.1 Blood

6.3.1.1 The Importance of Blood

Blood Composition:

Blood is a living tissue composed of four main components:

- **Plasma**: Yellowish liquid that transports blood cells and dissolved substances
- **Red blood cells (erythrocytes)**: Cells that lack a nucleus and transport oxygen
- **White blood cells (leukocytes)**: Larger cells that have a defense function
- **Blood platelets (thrombocytes)**: Cell fragments responsible for blood clotting

Blood Formation and Color:

The cellular elements of blood are formed in red bone marrow. Some white blood cells complete their development in the thymus and spleen. The red color of blood is due to hemoglobin, a substance present inside red blood cells.

Functions of Blood:

Blood performs three main functions:

1. **Transport**: Plasma transports nutrients and substances resulting from reactions that occur inside cells (metabolic waste). Red blood cells transport oxygen.
2. **Defense**: White blood cells fight invading microorganisms through two processes:
   - **Phagocytosis**: White blood cells surround and digest microorganisms
   - **Diapedesis**: White blood cells cross capillary walls to exit the blood
3. **Regulation**: Blood redistributes heat during circulation, maintaining body temperature.

Blood Analysis:

When analyzing blood values, it is important to consult reference values. Variations may indicate:

- **Low red blood cell values**: May indicate anemia
- **High white blood cell values**: May indicate an infection or inflammation
- **Altered platelet values**: May indicate clotting problems

Blood Groups:

Human blood can be classified into different groups by analyzing antigens present on the surface of red blood cells.

ABO System:

- Type A
- Type B
- Type AB
- Type O

Rh Factor (Rh positive or Rh negative):

Combining the ABO system with the Rh factor, we get 8 different blood groups.

Blood Compatibility:

- **Type AB**: Functions as "universal recipient" - can receive blood from any group
- **Type O**: Functions as "universal donor" - can donate blood to any group
- Rh negative individuals cannot receive Rh positive blood

6.3.2 Cardiovascular System

6.3.2.1 CIRCULATORY SYSTEM

Constitution of the cardiovascular system:

- **Blood:** responsible for transport of substances and for defense and regulation of the organism
- **Heart:** organ that propels blood
- **Blood vessels:** structures through which blood circulates

Heart:

Constitution:

- Right and left atria (receive blood through veins)
- Right and left ventricles (release blood through arteries)
- Atrioventricular valves (tricuspid and bicuspid)
- Sigmoid valves (semilunar)
- Septum (separates ventricles)
- Myocardium (muscle of contractions)
- Pericardium (membrane covers myocardium)

Cardiac cycle:

- **Diastole:** general relaxation - blood fills the atria and partially the ventricles
- **Atrial systole:** contraction of atria that pushes blood into ventricles
- **Ventricular systole:** contraction of ventricles that propels blood into arteries

Blood Vessels:

- **Arteries:** have thick, elastic walls, transport blood under high pressure
- **Veins:** have venous valves that prevent blood backflow, transport blood under low pressure
- **Capillaries:** small vessels that connect arteries to veins, allow exchange of substances

Types of Circulation:

Systemic Circulation: Transports oxygen-rich blood from the heart to all organs of the body, and brings oxygen-poor blood back to the heart.

Pulmonary Circulation: Transports oxygen-poor blood from the heart to the lungs, and brings oxygen-rich blood from the lungs back to the heart.

6.3.2.2 CARDIOVASCULAR SYSTEM HEALTH

Diseases of the Cardiovascular System:

- **Atherosclerosis:** accumulation of fatty deposits in artery walls
- **Thrombosis:** formation of clots (thrombi) inside blood vessels
- **Stroke:** interruption of blood flow to the brain
- **Coronary disease:** narrowing of arteries that irrigate the heart
- **Myocardial infarction:** death of heart muscle cells due to lack of oxygen
- **Valvulopathies:** diseases of heart valves

Contributions of Science:

- **Cardiac bypass:** surgery to bypass obstructed arteries
- **Heart valve replacement:** implantation of artificial or donor valves
- **Stent:** small tube inserted in arteries to keep them open
- **Heart transplant:** replacement of damaged heart

Preventive Measures:

To maintain cardiovascular system health, it is important to:

- Not smoke
- Healthy and balanced diet (reduce salt, saturated fats and cholesterol)
- Practice physical exercise regularly (at least 30 minutes per day)
- Maintain a healthy weight
- Control stress
- Reduce alcohol consumption
- Perform periodic blood pressure and cholesterol examinations
- Sleep enough (7-9 hours per night)
- Avoid excessive caffeine consumption

6.3.3 Lymphatic System

6.3.3.1 LYMPHATIC SYSTEM

Constitution of the lymphatic system:

- Lymph
- Lymphatic vessels
- Lymph nodes
- Lymphoid organs (tonsils, thymus, spleen)

Lymph - Constitution:

- White blood cells
- Plasma

How it is formed: Lymph originates when blood loses plasma and white blood cells when passing through capillaries.

Types of lymph:

- Interstitial lymph: existing in spaces between cells
- Circulating lymph: circulates in lymphatic vessels

Lymph Nodes - Function: Remove cellular debris and microorganisms from lymph, producing antibodies.

Lymphoid organs:

- Thymus: development and selection of white blood cells
- Tonsils: concentrate white blood cells and promote defense
- Spleen: controls, stores and destroys aged blood cells

Functions of the lymphatic system:

- Drainage: collection of fluids from interstitial spaces
- Defense: elimination of foreign bodies
- Nutrient absorption: fatty acids and fat-soluble vitamins
- Transport: delivery of substances to cells

6.3.3.2 LYMPHATIC SYSTEM HEALTH

Diseases:

- Tonsillitis: inflammation of tonsils with fever, sore throat and difficulty swallowing
- Lymphedema: excessive fluid accumulation causing swelling
- Lymphoma: cancer of lymphocytes with disordered growth

Preventive measures:

- Good personal hygiene
- Healthy diet
- Physical exercise
- Water intake (minimum one liter daily)
- Avoid chemical pollutants and radiation

6.4 Respiratory System

6.4.1 Respiratory System

6.4.1.1 Constitution of the Respiratory System

The respiratory system is constituted by the respiratory pathways and the lungs:

Respiratory Pathways:

- Nasal cavities
- Pharynx
- Larynx
- Trachea
- Bronchi
- Bronchioles

Functions of Structures:

Nasal cavities: Air enters and is filtered of dust and impurities, humidified and warmed.

Lungs: Organs responsible for gas exchange, where the processes of blood oxygenation and carbon dioxide elimination occur.

Pulmonary alveoli: Covered by a single layer of cells and irrigated by several blood capillaries. They are the structures where the main gas exchanges between air and blood occur.

Diaphragm: Muscle that functions as the main agent in pulmonary ventilation.

6.4.1.2 Pulmonary Ventilation

Pulmonary ventilation comprises two movements:

Inspiration: Movement of air entry into the lungs. The external intercostal muscles contract, the ribs elevate and the diaphragm contracts and lowers, increasing the volume of the thoracic cage and decreasing internal pressure, allowing air entry.

Expiration: Movement of air exit from the lungs. The external intercostal muscles relax, the ribs lower and the diaphragm relaxes and rises, decreasing the volume of the thoracic cage and increasing internal pressure, allowing air exit.

These movements are controlled by the nervous system, which detects changes in CO_2 and O_2 concentration in blood and cerebrospinal fluid.

6.4.1.3 Gas Exchanges

Pulmonary Hematosis: Process of gas exchange in pulmonary alveoli, where: - Oxygen present in air enters the blood - Carbon dioxide exits the blood to the alveoli - Diffusion occurs to areas where partial pressure is lower

Tissue Hematosis: Process of gas exchange between blood and tissue cells, where: - Oxygen exits the blood to cells - Carbon dioxide exits cells to blood - Diffusion occurs following the partial pressure gradient

6.4.1.4 Altitude Acclimatization

Acclimatization is the process in which the human organism adjusts to physical changes in the environment, such as high altitude.

When a person is at high altitude, where the partial pressure of oxygen is lower: - The organism increases respiratory frequency - Heart rate increases - The body stimulates the production of more red blood cells (erythrocytes) to increase oxygen transport capacity - Permanent residents at high altitudes naturally develop more red blood cells

6.4.1.5 Diseases of the Respiratory System

Pulmonary Emphysema: Disease in which pulmonary alveoli degrade and rupture, significantly reducing the gas exchange area. It is strongly associated with tobacco consumption.

Chronic Bronchitis: Chronic inflammation of airways that causes persistent cough. Main cause: tobacco consumption and air pollution.

Lung Cancer: Malignant disease that affects lung tissue. Main cause: tobacco consumption. Can develop in smokers and in people exposed to secondhand smoke.

6.4.1.6 Prevention Measures

To maintain respiratory system health, it is important to:

1. **Not smoke** - Avoid tobacco consumption and exposure to secondhand smoke
2. **Regular physical exercise** - Improve lung capacity and respiratory efficiency
3. **Airy environments with clean air** - Avoid polluted places
4. **Nasal breathing** - Nasal breathing humidifies and warms air, filtering impurities

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6.4.2 Basic Life Support

6.4.2.1 What to Do in Case of Cardiorespiratory Arrest

Definition of Cardiorespiratory Arrest (CRA):

Cardiorespiratory arrest represents "the sudden interruption of cardiac and respiratory functions."

Indicative Symptoms:

Signs include chest, abdomen or back pain, absence of respiratory movements and sounds, difficulty speaking, bluish coloration of lips and extremities (cyanosis), pallor and loss of consciousness.

Chain of Survival:

A sequence of procedures structured in links: 1. Call 112 – emergency recognition 2. Support – basic resuscitation maneuvers 3. Restore – defibrillation 4. Stabilize – post-resuscitation care

Basic Life Support (BLS):

Corresponds to links 1 and 2, ensuring ABC: airway, breathing and circulation.

Basic Life Support Algorithm:

1. Assess safety conditions
2. Assess state of consciousness
3. Open airway (if unconscious and no trauma)
4. Assess breathing using LLF (look, listen, feel)
5. Position in recovery position if breathing (no trauma)
6. Call 112 if not breathing
7. Request automated external defibrillation device (AED)
8. Begin chest compressions: 30 times (5-6 cm in adults; 15 times, 4 cm in children)
9. Perform ventilations: 2 for every 30 compressions
10. Maintain until arrival of help or normal breathing

Recovery Position Procedure:

1. Extend arms and legs
2. Remove dangerous objects
3. Bend nearest arm at right angle, palm up

4. Cross other arm over chest
5. Use crossed hand on face while bending knee
6. Rotate body keeping hip and knee at 90°
7. Tilt head back
8. Check breathing using LLF for up to 10 seconds

6.4.2.2 What to Do in Case of Airway Obstruction

Definition of Airway Obstruction (AO):

Consists of "an obstruction of a body (food, toy or other) in the respiratory pathways that prevents normal air flow." Mild obstructions allow air passage; severe ones completely prevent it.

Symptoms of Severe Obstruction:

Babies cannot cry; children and adults present acute sound on inspiration or total silence, cyanosis (bluish skin) and inability to speak or cough.

Procedure for Adults and Children Over One Year:

1. Apply up to five back blows (between shoulder blades)
2. If not resolved, apply up to five abdominal compressions (Heimlich maneuver)
3. Repeat cycle until resolved
4. If unconscious, begin basic life support

Procedure for Children Under One Year:

Back blows are performed on the lap or forearm with the child head down.

6.5 Excretory System

6.5.1 Excretory System: Urinary System

6.5.1.1 Functions of the Excretory System

The excretory system has as main functions the elimination of cellular metabolism waste and excess substances in the organism.

6.5.1.2 Constitution of the Excretory System

The excretory system is constituted by the respiratory system (eliminates carbon dioxide and water vapor), the urinary system and the skin.

The urinary system is the organ system that contributes most to the elimination of cellular metabolism waste and excess substances.

6.5.1.3 URINARY SYSTEM

Constitution of the urinary system:

The urinary system is constituted by the kidneys, ureters, bladder and urethra.

The kidneys are located in the posterior part of the abdominal cavity, one on each side of the spine. They are about 12 centimeters long, are bean-shaped, and have the function of filtering blood and forming urine. Blood reaches the kidneys through the renal arteries, branches of the aorta artery, and returns to circulation through the renal veins, which flow into the inferior vena cava.

The ureters are channels about 30 centimeters long and 5 millimeters in diameter that transport urine from the kidneys to the bladder.

The bladder is located in the pelvic cavity and is a hollow organ with elastic walls that stores urine until micturition (expulsion of urine to the outside). It is controlled by the nervous system and by two urethral sphincters.

The urethra is a channel about 4 centimeters in women and 20 centimeters in men that transports urine from the bladder to the outside.

6.5.1.4 Morphology of the Human Kidney

The human kidney is constituted by three regions: cortical zone (renal cortex), medullary zone (renal medulla) and renal pelvis.

The cortical zone is the outermost layer with a granular appearance.

The medullary zone has a striated appearance and is constituted by the pyramids of Malpighi.

The renal pelvis is the innermost zone of the kidney. It receives urine from the pyramids of Malpighi and conducts it to the ureter.

6.5.1.5 Functional and Structural Unit of the Kidney: The Nephron

The nephron is the functional and structural unit of the kidney, where urine formation occurs. It is found in the cortical and medullary zones.

Constitution of the nephron:

The nephron is constituted by the uriniferous tubule surrounded by blood vessels.

The uriniferous tubule is constituted by Bowman's capsule, the proximal convoluted tubule, the loop of Henle and the distal convoluted tubule.

The urine formed in the nephron is then collected by the collecting duct.

Blood vessels surrounding the nephron:

Blood reaches the nephron through the afferent arteriole, to the glomerulus of Malpighi (network of capillaries found inside Bowman's capsule). Then the efferent arteriole transports blood from

the glomerulus of Malpighi to the peritubular capillaries that surround the tubular part of the uriniferous tubule.

6.5.1.6 Stages of Urine Formation

Urine is formed in three stages: glomerular filtration, tubular reabsorption and tubular secretion.

Glomerular filtration consists of the passage of part of blood plasma, and micromolecules (water, mineral salts, glucose, amino acids, vitamins) and some cellular metabolism waste (urea and uric acid) present there, from the glomerulus of Malpighi to Bowman's capsule, originating a fluid called glomerular filtrate. Macromolecules, such as lipids and proteins, and formed elements cannot cross the blood capillaries due to their size.

Tubular reabsorption consists of the return of substances useful to the organism (such as water, mineral salts, glucose, amino acids and vitamins) from the glomerular filtrate to the blood. This passage of substances occurs in the region of the proximal and distal tubules and the loop of Henle, to the tubular capillaries. The amount of substances reabsorbed depends on their concentration in plasma, with excess not being reabsorbed.

Tubular secretion consists of the passage of substances harmful to the organism (nitrogenous waste such as urea, uric acid, creatinine and ammonia, drug residues and food additives) present in the peritubular capillaries to the uriniferous tubule, especially in the region of the distal convoluted tubule.

The formed urine is collected by the collecting duct, where some water reabsorption and secretion still occur, then passing to the renal pelvis.

6.5.1.7 Urine Composition

Urine is essentially constituted by water, mineral salts (such as chlorine, potassium and sodium), urea, uric acid and creatinine.

6.5.1.8 Urine Concentration

Abundant water intake and foods rich in water increase urine production and decrease its concentration.

Water loss through perspiration and excessive salt and meat consumption (urea production due to protein metabolism) increase urine concentration.

6.5.1.9 Regulatory Mechanism of the Urinary System

When the internal environment is very concentrated, the organism produces less and more concentrated urine.

When the internal environment is less concentrated, the organism produces more and less concentrated urine.

This urinary system mechanism process is a homeostasis process of the organism (organism adaptation to ensure its balance and proper functioning).

6.5.1.10 Behaviors that Promote Urinary System Health

Hygiene, hydration and healthy eating have fundamental roles in promoting urinary system health. We should also avoid holding urine and interrupting micturition.

6.5.1.11 Diseases of the Urinary System

Lithiasis results from the formation of renal calculi in the renal pelvis or ureters, by crystallization of excretion substances such as uric acid or calcium salts. When renal calculi grow they hinder renal function and cause very strong pain called renal colic. Causes for this disease are insufficient water intake, excessive consumption of foods rich in salt and kidney infections. To remove renal calculi, surgery or shock wave treatments that divide calculi into smaller parts facilitating their expulsion can be used.

Renal insufficiency consists of the loss of renal function by the kidney. Risk factors are smoking and diets rich in proteins. In the initial phase, treatment consists of medications and special diets, but if it reaches a critical level, treatments such as hemodialysis and renal transplant may be necessary.

6.5.1.12 Hemodialysis and Kidney Transplant

Hemodialysis consists of the passage of blood through an artificial semipermeable membrane outside the patient's body in order to remove toxic substances accumulated in the blood.

Kidney transplant is a surgery in which a healthy kidney is transferred to the patient. After the transplant, the patient must take immunosuppressive medications to avoid rejection of the new kidney. The diseased kidney is only removed if it is causing infection or hypertension.

6.5.2 Excretory System: Skin

6.5.2.1 THE SKIN

Skin Layers:

"The skin is formed by two layers: the epidermis and the dermis."

Epidermis: Outermost layer of skin. The superficial zone contains dead cells rich in keratin, a waterproof protein, that constantly renew. The innermost zone has living cells containing melanin, pigment that colors the skin and protects from ultraviolet radiation. Cells receive nutrients by diffusion, as there are no blood vessels in this layer.

Dermis: Innermost layer containing hair follicles, hair, sebaceous glands producing sebum, sweat glands producing sweat, blood vessels and nerve endings.

6.5.2.2 Structures Attached to Skin

The main attached structures include nails, hair, sebaceous glands and sweat glands.

Nails and Hair: Have protective function against external aggressions. When it is cold, erector muscles raise hairs creating a layer of thermal insulating air.

Sebaceous Glands: Produce sebum that exits through the same pore as hair. "Sebum protects the skin, making it more flexible, elastic and waterproof, and has antifungal and antibacterial properties."

Sweat Glands: Produce sweat containing water, sodium chloride, other salts, urea, uric acid and other substances. Sweat performs two functions: excretion (elimination of harmful substances) and thermoregulation (evaporation cools the body).

6.5.2.3 Functions of Skin

"The skin has as main functions the excretion of harmful and excess substances, body thermoregulation, and thanks to numerous nerve endings it also functions as a receptor of sensations (pain, heat, cold, pressure), thus defending the body against external or internal aggressions."

6.5.2.4 Care to Take with Skin

"For the skin to perform its functions effectively it must be kept clean and cared for."

Recommendations: - Clean skin every morning and evening - Wash face with warm water - Do not take very hot baths - Use cleaning and moisturizing products appropriate to skin type - Drink water daily - Use sunscreen on very sunny days - Do not pop pimples - Remove makeup before sleeping

6.5.2.5 Skin Cancer

"Skin cancer is the most serious disease that can affect the skin."

Main Cause: Excessive sun exposure (ultraviolet radiation).

Other Risk Factors: Exposure to carcinogenic agents such as tobacco smoke or X-rays.

Treatments: Surgery is usual. Topical chemotherapy, photodynamic therapy or radiotherapy may also be suggested.

6.6 Coordination and Regulation

6.6.1 Neuro-Hormonal System: Nervous System

6.6.1.1 Function of the Neuro-Hormonal System

The nervous system and hormonal system work jointly to "coordinate all other systems of the organism."

6.6.1.2 Constitution of the Nervous System

The nervous system is divided into: - **Central Nervous System (CNS):** brain and spinal cord - processes responses to stimuli - **Peripheral Nervous System (PNS):** nerves and nerve ganglia - establishes communication between CNS and rest of body

6.6.1.3 Basic Structural and Functional Unit: The Neuron

The neuron is the fundamental unit of the nervous system.

Neuron composed of:

- Cell body (with nucleus)
- Dendrites
- Axon (with terminal arborization)
- Myelin sheath (protective structure)

Structural sequence: Axon + myelin sheath → nerve fiber → nerve bundle → nerve

Nerve ganglion: zone of cell body concentration

6.6.1.4 Nerve Impulse

Communication occurs through "electrical signals called nerve impulses, caused by the unequal distribution of positive and negative ions inside and outside the neuron."

Direction of nerve impulse:

Path: dendrites → cell body → axon → terminal arborization

The terminal arborization releases "neurotransmitters" (chemical substances) captured by the next cell.

Synapse: region where neurotransmitter release occurs

6.6.1.5 Brain Constitution

The brain is protected by the skull and meninges, containing: - Cerebrum - Cerebellum - Medulla oblongata - Hypothalamus - Pituitary gland

Meninges: dura mater, arachnoid, pia mater

Cerebrospinal fluid: cushions shocks

6.6.1.6 Cerebrum Constitution

The cerebrum is "the largest organ of the brain" divided into: - Two hemispheres (left controls right side of body; right controls left side) - Longitudinal sulcus - Corpus callosum (unites hemispheres deeply)

External characteristics:

- Cerebral cortex
- Convolutions (protrusions)
- Sulci

6.6.1.7 Functions

Cerebrum:

- Voluntary motor activity
- Sensory activity (vision, hearing)
- Intellectual activity (memory, intelligence)

Cerebellum:

- Muscular coordination
- Body balance

Medulla oblongata:

- Involuntary vital movements (heartbeats, breathing, peristaltic movements)

6.6.1.8 Function of Spinal Cord

The spinal cord is protected by meninges and vertebrae. It conducts information between organs and brain and "determines involuntary acts."

6.6.1.9 Types of Nerves

By origin:

- **Cranial nerves:** connect brain to head organs (muscles, eyes, ears, nose, tongue)
- **Spinal nerves:** connect spinal cord to trunk and limb organs

By function:

- **Sensory nerves:** transport information from receptors to nerve centers
- **Motor nerves:** transport information from nerve centers to effector organs

6.6.1.10 Reflex Arc

Voluntary act: depends on will, controlled by cerebrum

Reflex/involuntary act: does not depend on will, spinal cord is response center

Reflex arc path: Stimulus → receptor organ → sensory neurons → association neurons (spinal cord) → motor neurons → effector organ

6.6.1.11 Somatic and Autonomic Nervous System

Peripheral Nervous System subdivides into:

Somatic Nervous System: voluntary movements

Autonomic Nervous System: involuntary movements - **Parasympathetic:** energy conservation, rest - **Sympathetic:** prepares for emergencies (increases heart rate, dilates bronchi, inhibits micturition)

6.6.1.12 Diseases that Affect the Nervous System

Multiple Sclerosis:

- Disabling, degenerative, progressive disease
- Inflammation/damage to myelin sheath
- Affects: vision, speech, movement, balance, sensitivity
- No cure; medications and physiotherapy relieve symptoms

Spinal Cord Injuries:

- Loss of sensitivity and mobility below injury
- Possible loss of autonomous breathing

Meningitis:

- Inflammation of meninges by viral/bacterial infection
- Can cause death or serious sequelae
- Prevention: vaccination against bacterial form

6.6.1.13 Measures that Promote Nervous System Health

Necessary to maintain: - Healthy lifestyle - Intellectual activity - Stress reduction - Sufficient rest

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6.6.2 Neuro-Hormonal System: Hormonal System

6.6.2.1 Constitution of the Hormonal System

"The hormonal system is constituted by endocrine glands, organs that produce hormones and release them into the blood."

6.6.2.2 Hormones and Target Cells

"Hormones are substances that have the capacity to stimulate or inhibit the activity of certain cells – target cells." These target cells have receptors that identify a specific type of hormone.

6.6.2.3 Endocrine Glands and Functions

- **Pineal gland**: produces melatonin, responsible for regulating biological rhythms, such as activity/rest cycles
- **Pituitary gland**: produces growth hormone, which accelerates cell division and growth

- **Hypothalamus**: produces stimulating or inhibiting hormones of the pituitary gland
- **Thyroid**: produces calcitonin, which regulates calcium in blood and its fixation in bones
- **Adrenal glands**: produce adrenaline, which prepares the organism for stress situations
- **Islets of Langerhans** (pancreas): produce insulin, which acts in glucose metabolism
- **Ovaries** (female): produce estrogens (female sexual characteristics) and progesterone (sexual cycle regulation)
- **Placenta** (pregnancy): produces hormones that replace those of the ovary
- **Testes** (male): produce testosterone (male sexual characteristics and sperm production)

6.6.2.4 Link Between Hormonal System and Nervous System

"The nervous and hormonal systems function jointly to control physiological processes and maintenance of homeostasis." The hypothalamus intervenes in nervous and hormonal regulation, ensuring the link between the two types of regulation.

6.6.2.5 Diseases that Affect the Hormonal System

Type I Diabetes: pancreas cells produce little or no insulin. Treatment consists of taking insulin several times a day. There are insulin pumps that automatically perform glycemia analysis and its administration.

Hypothyroidism: the thyroid produces hormones below what is necessary, and can cause dementia and depression. The most frequent cause is iodine deficiency, which causes enlargement of the thyroid (goiter). Diet rich in foods of marine origin (rich in iodine) prevents this disease.

Pituitary Hypofunction: insufficient production of growth hormone. Has no serious consequences in adults, but in children reduces growth velocity and can cause dwarfism.

Pituitary Hyperfunction: can lead to exaggerated growth (gigantism). If it occurs in adults, it can lead to alterations and enlargement of organs such as heart, liver and spleen.

6.6.2.6 Measures that Promote Hormonal System Health

Special attention to diet because many hormones are produced from food substances. Control of cholesterol and hypertension, as well as practice of physical exercise.

6.7 Reproduction and Heredity

6.7.1 Reproductive System

6.7.1.1 Adolescence and Sexual Characteristics

Adolescence: Period between 10-19 years involving biological, intellectual, emotional, sexual and social changes

Puberty: Period of biological changes that prepares the body for reproduction, marking the beginning of adolescence

Primary sexual characteristics: Reproductive organs present since embryonic development

Secondary sexual characteristics: External morphological characteristics that appear during puberty

Gonads: Testes (male) and ovaries (female); produce gametes and sex hormones

Gametes: Sex cells—sperm (male) and oocytes (female)—that fuse to form a zygote

6.7.1.2 Male Reproductive System

The male reproductive system is constituted by the testes (gonads), epididymis, vas deferens and related structures, producing sperm and testosterone.

Key organs and functions:

- **Testes:** Produce sperm and testosterone; protected by scrotum
- **Epididymis:** Tubular organs where sperm matures and is stored
- **Vas deferens:** Tubes that transport sperm to urethra
- **Urethra:** Channel that conducts sperm out of body
- **Penis:** Organ for introducing sperm into vagina
- **Seminal vesicles:** Produce seminal fluid rich in nutrients
- **Prostate:** Produces prostatic fluid; neutralizes vaginal acidity
- **Cowper's glands:** Produce secretion that cleans and lubricates urethra

6.7.1.3 Female Reproductive System

The female reproductive system is constituted by the ovaries (gonads), fallopian tubes, uterus and vagina.

Key organs and functions:

- **Ovaries:** Produce oocytes and hormones (estrogen, progesterone)
- **Fallopian tubes:** Location where fertilization occurs
- **Uterus:** Muscular organ lined with endometrium where embryo implants and develops
- **Vagina:** Muscular channel that receives sperm and functions as birth canal
- **Vulva:** External genitalia including clitoris and labia

- **Bartholin's glands:** Produce lubricating mucus

6.7.1.4 Spermatogenesis

Spermatogenesis is the process of formation, development and maturation of male gametes that occurs in the testes from puberty throughout life.

It occurs within the seminiferous tubules of the testes. Cells divide continuously along the tubule walls. Results in mature sperm with distinct regions:

Sperm structure:

- **Head:** Contains nucleus with genetic information (chromosomes); covered by acrosome with enzymes
- **Middle segment:** Contains mitochondria producing ATP energy
- **Tail:** Allows sperm movement

6.7.1.5 Oogenesis

Oogenesis (or ovogenesis) is the process of formation of female gametes from germinal cells in the ovaries, beginning during fetal development and continuing from puberty to menopause (approximately at age 50).

Timeline:

- Begins during embryonic development; suspended until puberty
- From menarche (first menstruation), follicles develop monthly
- One follicle completes oogenesis monthly, releasing an oocyte through ovulation
- Remaining follicular tissue forms corpus luteum
- Cycle repeats approximately every 28 days

Oocyte characteristics:

- Contains half the chromosomes of the original germinal cell
- Abundant cytoplasm rich in reserve substances
- Size: 250,000 times larger than sperm
- Viability: 24-72 hours after ovulation if not fertilized
- Sperm viability: 3-5 days in female body

6.7.1.6 Female Menstrual Cycle

The female sexual cycle, or menstrual cycle, is the set of cyclical phenomena (on average every 28 days) with coordinated changes in ovaries and uterus.

Stages of Ovarian Cycle:

Follicular phase (first 14 days): - Production of estrogen by follicular cells - Growth and maturation of a follicle containing an oocyte

Ovulation (14th day): - Release of oocyte to fallopian tube

Luteal phase:

- Formation of corpus luteum producing estrogen and progesterone
- Persists during early pregnancy if fertilization occurs; degenerates if not

Stages of Uterine Cycle:

Menstrual phase (approximately 5 days): - Shedding of superficial layer of endometrium with hemorrhage (menstruation)

Proliferative phase (until ovulation on day 14): - Endometrium reconstruction begins due to estrogen increase

Secretory phase (after ovulation): - Endometrium reaches maximum thickness due to progesterone increase - Prepares to receive embryo if fertilization occurs - Returns to menstrual phase if fertilization does not occur

6.7.1.7 Fertile Period

The fertile period is the time interval of the menstrual cycle when women have the highest probability of conception.

Calculation:

- Ovulation occurs 14 days before next menstruation
- Oocyte viable for 24 hours
- Sperm viable up to 5 days after sexual intercourse
- **Fertile window: 5 days before to 3 days after ovulation**

6.7.1.8 Hormonal Regulation

The functioning of the reproductive system is regulated by the nervous and hormonal systems. The hypothalamus stimulates the pituitary gland to produce two sex hormones:

- **FSH** (follicle-stimulating hormone)
- **LH** (luteinizing hormone)

These hormones are released into the bloodstream to reach the testes and ovaries.

6.7.1.9 Fertilization and Development

Fertilization: Fusion of sperm with ovum, forming a single cell—the egg or zygote

Sequence of events:

1. Formation of morula
2. Nidation (implantation of blastocyst in endometrium)
3. Formation of placenta, umbilical cord and amniotic sac

4. Embryo → Fetus (at 3 months when human form develops, called fetus)

Structures:

- **Placenta:** Connects embryo to mother's blood supply
- **Umbilical cord:** Connects fetus to placenta
- **Amniotic sac:** Protects embryo from shocks and temperature variations

6.7.1.10 Stages of Childbirth

Dilation:

- Uterine contractions occur
- Cervix dilates
- Amniotic sac breaks

Expulsion of fetus:

- Fetus passes through cervix and vagina
- Baby breathes independently for first time
- Umbilical cord is cut

Expulsion of placenta:

- Remaining umbilical cord and placenta exit
- Endometrial debris is expelled
- Occurs naturally approximately 1 minute after fetus delivery

6.7.1.11 Importance of Breastfeeding

Breast milk contains all essential nutrients in quantities adequate for infant digestion and development.

6.7.2 Heredity

6.7.2.1 Learning Objectives

According to the Portuguese curriculum, students should understand:

1. Distinguish between concepts of genetics and heredity
2. Historical evolution of genetics, including contributions of Gregor Mendel and Thomas Morgan
3. Cellular structures that contain genetic material
4. Relationships between hereditary factors and genetic information
5. Calculate probability of some hereditary characteristics (autosomal and heterosomal)
6. How sexual reproduction affects intraspecific diversity

7. Three applications of genetics in society
8. Bioethical problems related to new genetic applications

Content adapted from: O Bichinho do Saber (www.obichinhodosaber.com)